

Choosing variables

A delivery fee depends on distance traveled.

Which quantity should be the input, and what units should each variable use?

Building a cost model

A workshop charges a \$35 registration fee plus \$8 per session. Let n count sessions.

What function gives the total cost?

Interpreting a model

A tank initially contains 120 liters and drains at 6 liters per minute.

$$V(t) = 120 - 6t$$

What does each parameter represent?

The physical domain

The tank begins with 120 liters and loses 6 liters per minute until empty.

What time interval belongs in the model domain?

Predicting an output

A printer charges \$24 setup plus \$0.15 per page.

$$C(n) = 24 + 0.15n$$

What is the cost of printing 160 pages?

Finding an input

A printer charges \$24 setup plus \$0.15 per page.

$$C(n) = 24 + 0.15n$$

How many pages can be printed for exactly \$54?

Building from two observations

A cooler warms at a constant rate: 8 degrees Celsius at hour 1 and 14 degrees at hour 4.

What temperature function $T(t)$ fits these observations?

Changing the reference time

A balance is \$260 three weeks after saving begins and grows by \$40 each week.

What was the balance when saving began?

Comparing plans

Monthly costs in dollars depend on the number n of visits.

$$A(n) = 20 + 6n, \quad B(n) = 44 + 3n$$

For how many visits do the plans cost the same?

Choosing a plan

Monthly costs in dollars depend on the number n of visits.

$$A(n) = 20 + 6n, \quad B(n) = 44 + 3n$$

For which nonnegative integer values of n is plan B cheaper?

Break-even

A seller receives \$15 per item. Fixed costs are \$180 and production costs are \$6 per item.

How many items must be sold for revenue to equal cost?

A linear conversion

A measuring device displays 14 units for 2 cm and 38 units for 8 cm. Assume a linear conversion.

What formula converts centimeters to displayed units?

Exact model or approximation?

Observed pairs are $(0, 10)$, $(1, 13)$, $(2, 17)$, and $(3, 19)$.

Do all four points lie on one line?

Checking a prediction

A linear model for a draining tank predicts a negative volume after 30 minutes.

What should this prediction tell us about using the model?